

Generalized linear models

Iterative solution of the estimation equation

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Iterative solution of the estimation equation - Contents

Solving the estimating equation

Newton-Raphson method

Fisher's scoring method

Iteratively Weighted Least Squares

Solving the estimating equation

$$\hat{\beta} = (\underline{X^T X})^{-1} \underline{X^T y}$$

- Solving the estimating equation

$$\begin{aligned} \mathbf{s}(\beta) &= \mathbf{0} \\ \mathbf{s}\left((\beta_0, \beta_1, \dots, \beta_p)^T\right) &= \mathbf{0} \end{aligned}$$

only in special
case 3, e.g.
Gaussian

is not explicitly possible for general GLMs.

- Instead, numerical algorithms are used for an approximate solution. The three most common methods are:
 - Newton-Raphson
 - Fisher-Scoring
 - IWLS (Iteratively weighted LS-estimation)

Solving the estimating equation

Iterative method

- ▶ Iterative method: starting from initial value $\hat{\beta}^{(0)}$, the estimator is successively improved

$$\hat{\beta}^{(0)} \xrightarrow{\text{improve}} \hat{\beta}^{(1)} \xrightarrow{\text{improve}} \hat{\beta}^{(2)} \xrightarrow{\dots \text{improve}} \dots \xrightarrow{\dots \text{improve}} \hat{\beta}^{(k)}$$

- ▶ Stop, if the relative improvement drops under a given threshold ε :

$$\frac{\|\hat{\beta}^{(k+1)} - \hat{\beta}^{(k)}\|}{\|\hat{\beta}^{(k)}\|} < \varepsilon$$

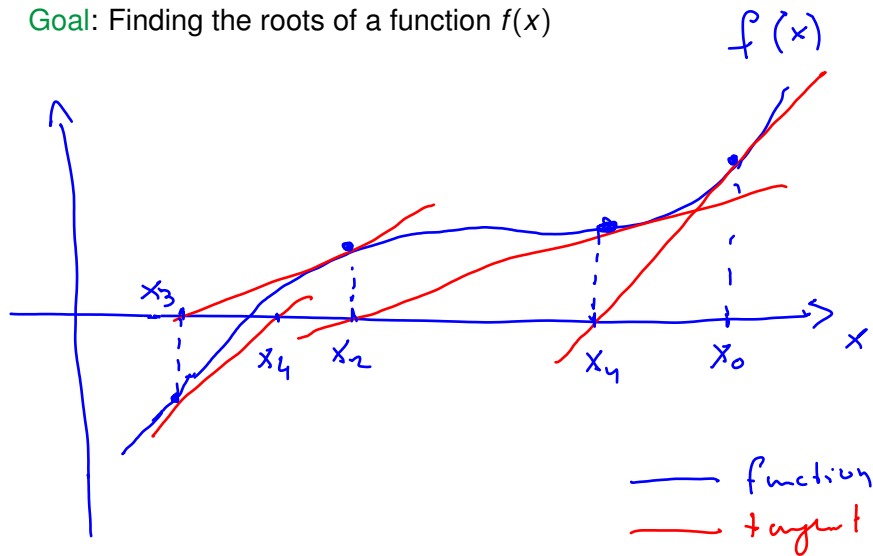
$\|\cdot\|_2$ each norm
"

→ Use $\hat{\beta}^{(k)}$ as estimator for β
 $\hat{\beta}^{(k+1)}$

$$\sqrt{\sum_{j=0}^p (\hat{\beta}_j^{(k+1)} - \hat{\beta}_j^{(k)})^2}$$

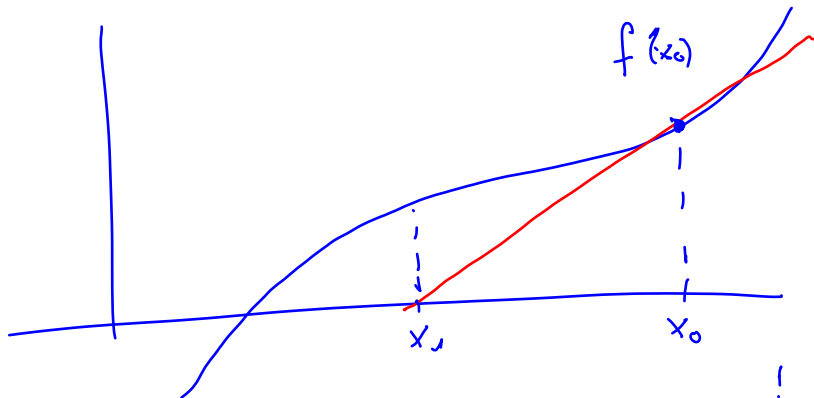
Newton-Raphson method

Goal: Finding the roots of a function $f(x)$



Newton-Raphson method

Inclusive formulas



tangent:
$$g(x^{(k+1)}) = f(x^{(k)}) + f'(x^{(k)})(x^{(k+1)} - x^{(k)}) \stackrel{!}{=} 0$$

$$\Leftrightarrow x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$$

linear Taylor approximation

Newton-Raphson method

Newton-Raphson method

Goal: Finding the roots of $\mathbf{s}(\beta)$

Init: Set $\hat{\beta}^{(0)}$

Iter: For $k = 1, 2, \dots$ iterate

$$\begin{aligned}\hat{\beta}^{(k+1)} &= \hat{\beta}^{(k)} - \left(\frac{\partial \mathbf{s}(\hat{\beta}^{(k)})}{\partial \beta} \right)^{-1} \mathbf{s}(\hat{\beta}^{(k)}) \\ &= \hat{\beta}^{(k)} + \mathbf{F}_{\text{obs}} \left(\beta^{(k)} \right)^{-1} \mathbf{s}(\hat{\beta}^{(k)})\end{aligned}$$

Stop: If convergence criterion is accomplished

Newton-Raphson method

Annotations:

- ▶ How to choose initial vector $\hat{\beta}^{(0)}$?

e.g. via LS-estimator

- ▶ Which requirements have to be fulfilled to make the method work?

- $\nabla_{\text{obs}}(\hat{\beta}^{(k)})$ invertible $(p+1 \times p+1)$
- no linear dependencies between columns of $\underline{X} \Rightarrow$ full column rank
- typically convergence achieved

Fisher's scoring method



Fisher's scoring

Goal: Finding the roots of $\mathbf{s}(\beta)$

Init: Set $\hat{\beta}^{(0)}$

Iter: For $k = 1, 2, \dots$ iterate

$$\hat{\beta}^{(k+1)} = \hat{\beta}^{(k)} + \mathbf{F}(\hat{\beta}^{(k)})^{-1} \mathbf{s}(\hat{\beta}^{(k)})$$

Stop: If convergence criterion is accomplished

*replaced observed
Fisher
matrix*

by the expected one

*→ classical method
for GLMs*

Iteratively Weighted Least Squares (IWLS)

$$LS: \hat{\beta} = (X^T X)^{-1} X^T y$$

IWLS

$$\hat{\beta}^{(k+1)} = (X^T W(\hat{\beta}^{(k)}) X)^{-1} X^T W(\hat{\beta}^{(k)}) \tilde{\eta}(\hat{\beta}^{(k)})$$

and

$$\tilde{\eta}(\hat{\beta}^{(k)}) = X \hat{\beta}^{(k)} + D^{-1}(\hat{\beta}^{(k)}) (y - \mu(\hat{\beta}^{(k)}))$$

working / pseudo response

Interpretation: LS-estimator with weights $\underline{w}(\hat{\beta}^{(k)})$
and $\underline{\tilde{y}}$ instead of y